

Machine Learning (CSE543/ECE543)

Quiz - 1 (20 Marks) (Duration 60 min)

- Beginner

1 point each

1. Logistic Regression is used for classification, while Linear Regression is used for approximation. What do we mean by Regression here, though both the models are doing a different task?
2. What is the difference between a model and an algorithm? Give one example.
3. The logistic S-curve is non-linear. Why do we still call the Logistic Regression a Linear model?
4. What is difference between simple and multiple regression?
5. What is the difference between knowledge-based learning (KBL) and feedback-based learning (FBL)? Give an example of an FBL model that is also a KBL model.
6. What is PET in the context of ML?
7. In the context of multivariate linear regression, prove that OLS is an unbiased Estimator.
8. Raju is working on a linear regression task with OLS where a number of predictors ($p > n$) a number of observations, and he is not getting results. Comment on why this setting will not work.

- Intermediate

2 points each

1. What are the two broad ways to reduce/avoid overfitting (Hint: Think about function approx. in ERM framework)?
2. What are the two important factors determining the generalization ability of an ML model? Graphically show when a model will fail to generalize.
3. Explain in detail the difference between MLE, MCLE and MAP methods.
4. Consider a linear regression model where the errors ϵ are not independent but rather follow a correlation structure, say $Cov(\epsilon_i, \epsilon_j) = \rho^{|i-j|}$ for some $\rho \in [0, 1]$. Derive the generalized least squares (GLS) estimator for this model.
5. Consider the estimate $MAP = f_{Y|X}(y|x)f_X(x)$. Let X be a continuous random variable with the following PDF $f_X(x) = [2x \ (0 < x < 1); 0 \text{ otherwise}]$. Also suppose

$$P_{Y|X}(y|x) = x(1-x)^{y-1}, \quad \text{for } y = 1, 2, \dots \text{ (Geometric dist.)}$$

Find the MAP estimate of X given $Y = 3$

- Advanced

1. Derive logistic regression when the assumption is made that the target variable originates from a Gaussian distribution. 2 points

- Bonus

1. A consistent estimator has the property that as the number of data points (which are used to estimate the parameters) increase, i.e. $N \rightarrow \infty$, the estimates converge in probability to the true parameters. Show that an unbiased estimator is not always consistent. Further, show that the OLS estimator is always consistent. You may use an example/counterexample to show the result. 2 points